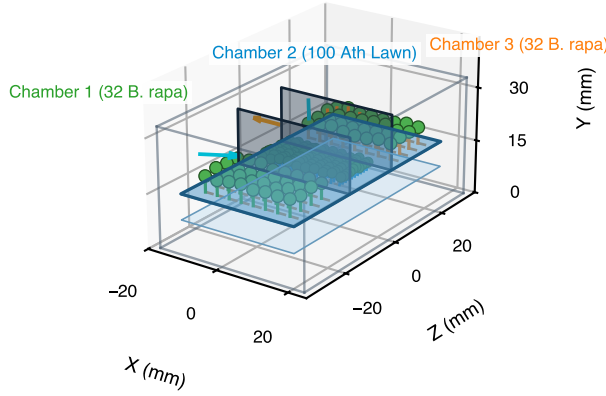
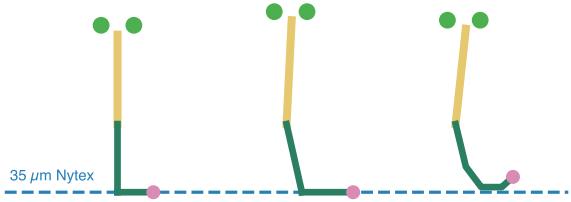
a LEAF GBC1 3D Chamber Digital Twin: 164-Seed Community & Push CFD Streamlines



b Lesson 1: Gravitational Regimes & Statolith Steering

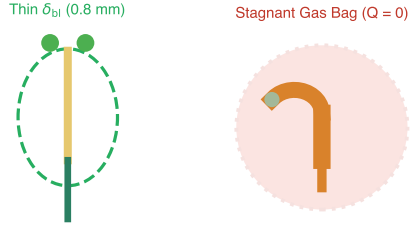


1.0g Earth
- Rapid sedimentation
- Uniaxial downward
- High statolith flux

0.166g Moon
- 83.4% lower buoyancy
- Delayed turning (35°)
- Broad Nytex contact

0g Microgravity
- No sedimentation
- Autotropic waving
- Stochastic curling

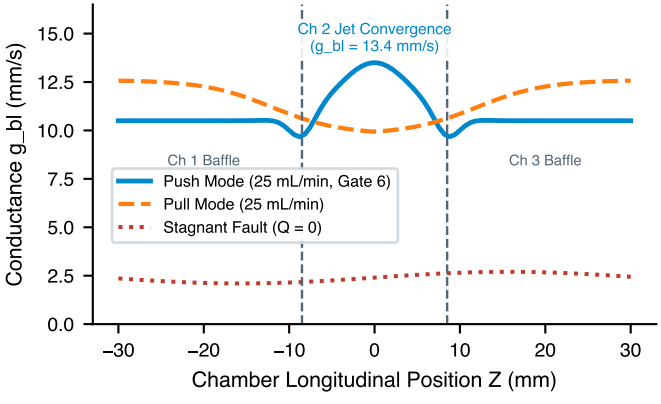
c Lesson 2: Airflow & Ethylene Triple Response



Active Push (25 mL/min)
- C_2H_4 < 0.1 ppb (purged)
- g_{bl} = 13.4 mm/s
- Slender, expanded shoot

Stagnant Fault (Pump Off)
- C_2H_4 = 136.5 ppb (trapped)
- g_{bl} < 3.0 mm/s
- +85% Radial Swelling

d Boundary-Layer Conductance & Airflow Channelling



e Organ Morphometrics: Gravity × Ventilation Matrix

